Лабораторная работа № 5

Эмуляция и измерение задержек в глобальных сетях

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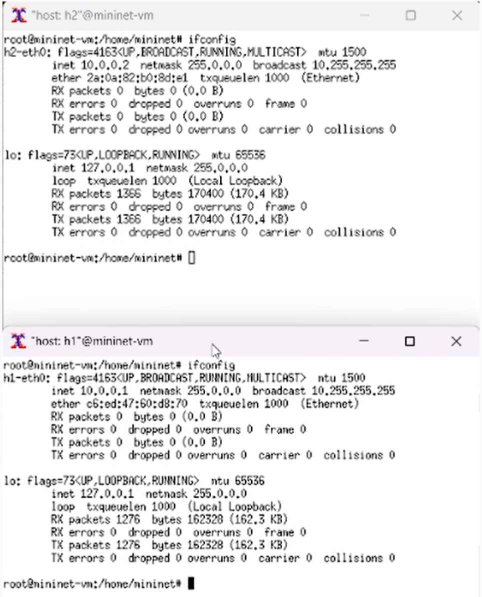
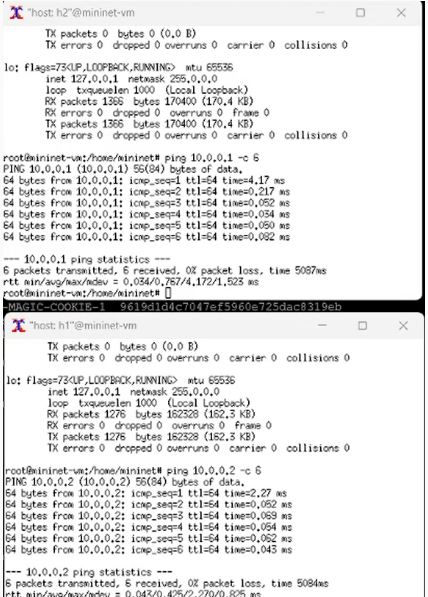
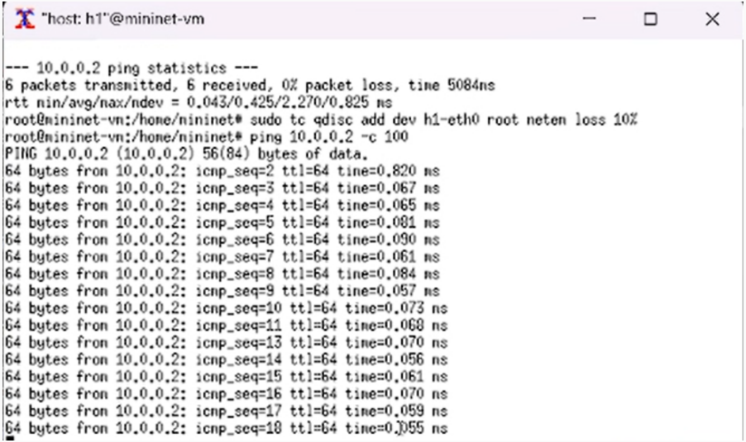
Цель работы

Основной целью работы является получение навыков проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных. Эти параметры влияют на производительность протоколов и сетей.

Выполнение лабораторной работы



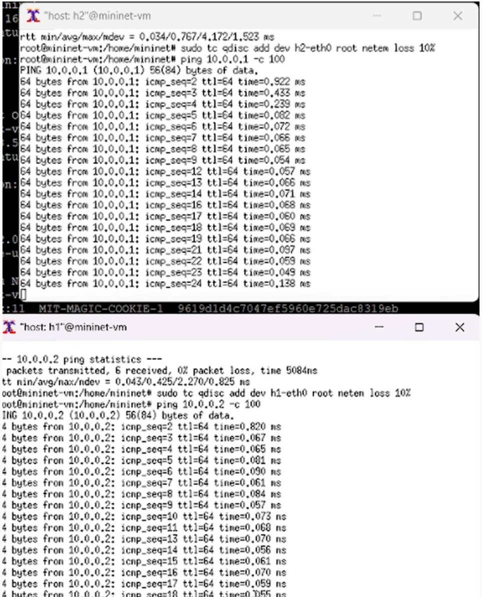

"host: h2"@mininet-vm
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
loop txqueuelen 1000 (Local Loopback)
RX packets 1366 bytes 170400 (170.4 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 1366 bytes 170400 (170.4 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
root@mininet-vm:/home/mininet# ping 10.0.0.1 -c 6
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=4.17 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.217 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.052 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.034 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.050 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.082 ms
--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5087ms
rtt min/avg/max/mdev = 0.034/0.767/4.172/1.523 ms
root@mininet-vm:/home/mininet#
-MAGIC-COOKIE-1 9619d1d4c7047ef5960e725dac8319eb
"host: h1"@mininet-vm
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
loop txqueuelen 1000 (Local Loopback)
RX packets 1276 bytes 162328 (162.3 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 1276 bytes 162328 (162.3 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 6
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.27 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.069 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.054 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.043 ms
--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5084ms



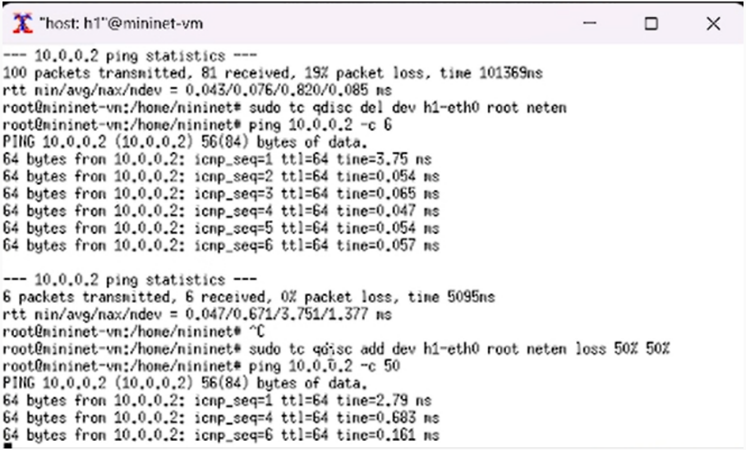
"host: h1"@mininet-vm

```
--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5084ms
rtt min/avg/max/mdev = 0.043/0.425/2.270/0.825 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 10%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 100
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.820 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.067 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.065 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.081 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.090 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.084 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.057 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.073 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.068 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.070 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.056 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.070 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.055 ms
```

Рис. 3: Добавление потери пакетов


"host: h2"@mininet-vm
rtt min/avg/max/mdev = 0.034/0.767/4.172/1.523 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h2-eth0 root netem loss 10%
root@mininet-vm:/home/mininet# ping 10.0.0.1 -c 100
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.922 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.433 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.239 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.082 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.072 ms
64 bytes from 10.0.0.1: icmp_seq=7 ttl=64 time=0.066 ms
64 bytes from 10.0.0.1: icmp_seq=8 ttl=64 time=0.065 ms
64 bytes from 10.0.0.1: icmp_seq=9 ttl=64 time=0.054 ms
64 bytes from 10.0.0.1: icmp_seq=12 ttl=64 time=0.057 ms
64 bytes from 10.0.0.1: icmp_seq=13 ttl=64 time=0.066 ms
64 bytes from 10.0.0.1: icmp_seq=14 ttl=64 time=0.071 ms
64 bytes from 10.0.0.1: icmp_seq=16 ttl=64 time=0.068 ms
64 bytes from 10.0.0.1: icmp_seq=17 ttl=64 time=0.060 ms
64 bytes from 10.0.0.1: icmp_seq=18 ttl=64 time=0.069 ms
64 bytes from 10.0.0.1: icmp_seq=19 ttl=64 time=0.066 ms
64 bytes from 10.0.0.1: icmp_seq=21 ttl=64 time=0.097 ms
64 bytes from 10.0.0.1: icmp_seq=22 ttl=64 time=0.059 ms
64 bytes from 10.0.0.1: icmp_seq=23 ttl=64 time=0.049 ms
64 bytes from 10.0.0.1: icmp_seq=24 ttl=64 time=0.138 ms
:11 MIT-MAGIC-COOKIE-1 9619d1d4c7047ef5960e725dac8319eb
"host: h1"@mininet-vm
-- 10.0.0.2 ping statistics ---
packets transmitted, 6 received, 0% packet loss, time 5084ms
tt min/avg/max/mdev = 0.043/0.425/2.270/0.825 ms
oot@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 10%
oot@mininet-vm:/home/mininet# ping 10.0.0.2 -c 100
ING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
4 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.820 ms
4 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.067 ms
4 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.065 ms
4 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.081 ms
4 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.090 ms
4 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.061 ms
4 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.084 ms
4 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.057 ms
4 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.073 ms
4 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.068 ms
4 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.070 ms
4 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.056 ms
4 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.061 ms
4 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.070 ms
4 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.059 ms

Выполнение лабораторной работы

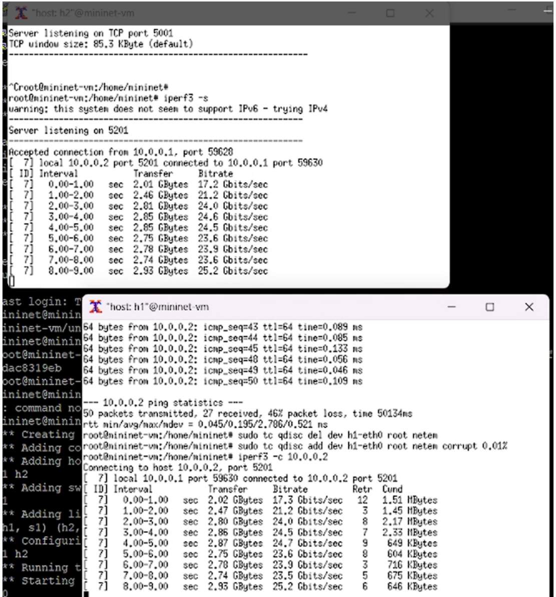


```
"host: h1"@mininet-vm
--- 10.0.0.2 ping statistics ---
100 packets transmitted, 81 received, 19% packet loss, time 101369ms
rtt min/avg/max/mdev = 0.043/0.076/0.820/0.085 ms
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 6
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.75 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.054 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.065 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.054 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.057 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5095ms
rtt min/avg/max/mdev = 0.047/0.671/3.751/1.377 ms
root@mininet-vm:/home/mininet# ^C
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 50% 50%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 50
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.79 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.683 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.161 ms
```

Рис. 5: Добавление значения корреляции для потери пакетов

Выполнение лабораторной работы



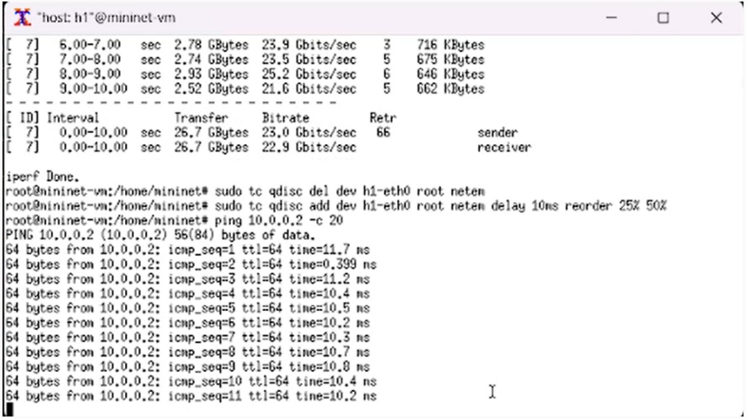


Рис. 7: Добавление переупорядочивания пакетов

```
"host: h1"@mininet-vm
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=10.3 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19084ms
rtt min/avg/max/mdev = 0.083/9.558/11.710/3.133 ms
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem duplicate 50%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 20
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.81 ms
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.82 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.640 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.14 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.183 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.086 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.086 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.132 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.132 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.125 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.126 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.057 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.058 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.061 ms (DUP!)
```

Рис. 8: Добавление дублирования пакетов

```
Output: ping.dat
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info
import time

def emptyNet():

        "Create an empty network and add nodes to it."

        net = Mininet( controller=Controller, waitConnected=True )
        info( '*** Adding controller\n' )
        net.addController( 'c0' )

info( '*** Adding hosts\n' )
h1 = net.addHost( 'h1', ip='10.0.0.1' )
h2 = net.addHost( 'h2', ip='10.0.0.2' )

info( '*** Adding switch\n' )
s1 = net.addSwitch( 's1' )

info( '*** Creating links\n' )
net.addLink( h1, s1 )
net.addLink( h2, s1 )

info( '*** Starting network\n')
net.start()

info( '*** Set delay\n')
h1.cmdPrint( 'tc qdisc add dev h1-eth0 root netem loss 10%' )
h2.cmdPrint( 'tc qdisc add dev h2-eth0 root netem loss 10%' )

time.sleep(10) # Wait 10 seconds

info( '*** Ping\n')
h1.cmdPrint( 'ping -c 100', h2.IP(), '| grep "time=" |
```

Выполнение лабораторной работы

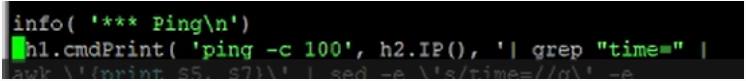


Рис. 10: Редактирование сркипта

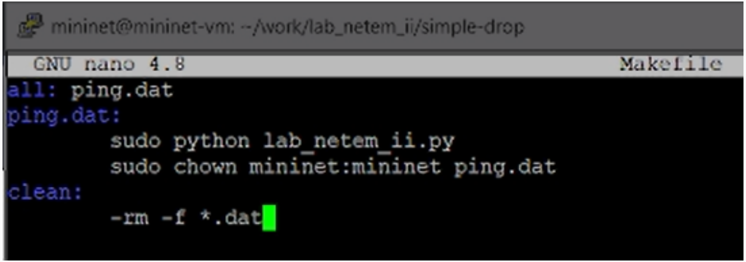


Рис. 11: Makefile

Выполнение лабораторной работы

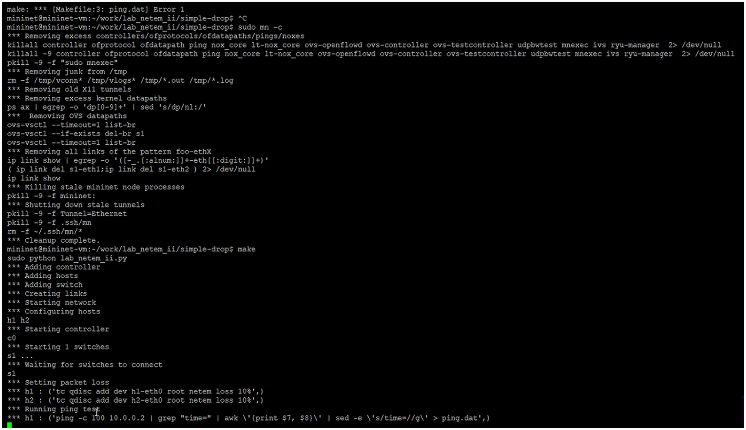


Рис. 12: Проведение эксперимента

Выводы

В результате выполнения данной лабораторной работы я получила навыки проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных.